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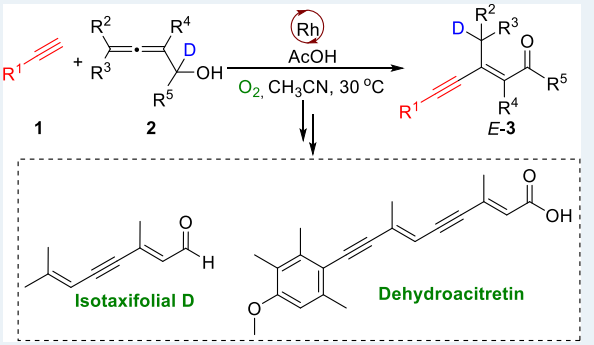
Rh-Catalyzed Aerobic *E*-Selective Linear Coupling of Terminal Alkynes with 2,3-Allenols

Yaqi Shi,[§] Xiaoyan Wu,[§] Chunling Fu, and Shengming Ma*

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ABSTRACT: Herein, a rhodium-catalyzed highly chemo-, regio-, and stereo-selective linear cross coupling reaction of terminal alkynes with readily available 2,3-allenols afforded the synthetically and biologically versatile 3-alkyn-2(*E*)-enones/enals. The reaction exhibits a perfect match of the reactivities for these two classes of compounds since the well-known self-reacting products of both reactants were not formed. The reaction enjoys a wide substrate scope under ambient conditions. Furthermore, the synthetic potentials have been demonstrated by utilizing this reaction as a pivotal step: a very concise two-step total synthesis of naturally occurring isotaxifolial D has been achieved; an efficient synthesis of dehydroacitretin by applying this reaction two times has also been executed. Through mechanistic studies including D-labeling experiments, it is proposed that the reaction proceeded via terminal alkynylmetalation, stereodefined insertion providing a 2-alkynyl π-allylic intermediate, which would undergo C−C single bond rotation, intramolecular ligand exchange, β−H elimination, and reductive elimination resulting in 1,4-H delivery to afford the final enynes. The reaction circumvents the use of a stoichiometric amount of base under the catalysis of a single Rh catalyst since terminal alkynylmetalation is base free, and the catalyst is regenerated by aerobic oxidation in the presence of proton.



KEYWORDS: *rhodium catalysis, 1,4-H delivery, enynes synthesis, allenes, alkynes, E-selectivity*

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INTRODUCTION

Transition metal-catalyzed reactions have constantly been providing highly efficient methodologies for the efficient formation of covalent bonds. With the fine-tuning of the ligand structure and metal, the regio-, chemo-, and stereo-selectivity may be precisely controlled. Terminal alkynes make up an important class of compounds for such reactions in organic synthesis. For example, the well-known Sonogashira coupling of terminal alkynes with 1-alkenyl halides utilizes the highly reactive C−X bond to form the sp C-sp^2 C bonds under the co-catalysis of Pd and Cu(I).[1] On the other hand, allenes have also been becoming one of the most important classes of organic compounds.[2] Nowadays, the allenation of terminal alkynes (ATA reaction) has made mono-, di-, and tri-substituted allenes readily available.[2d] The reactions of terminal alkynes with allenes have merely been sporadically studied. Of our interest is the formation of 1-alkynyl π-allylic metallic species **A**, which may be converted to enynes **A**′ and/or the regioisomeric enynes **A**″ upon further trapping, although the formations of non-conjugated alken-4-yn-2-yl metallic species **B** and **C** were also observed (Figure 1a).[3] Beyond terminal alkynes, decarboxylative alkynylation of allenes with alkynyl carboxylic acids[4] and transition metal-catalyzed reactions of haloalkynes with allenes[5] had also been developed. In 2005, Alcaide and Almendros reported the reaction of terminal alkynes with 2,3-allenols formed π-allylic metallic species **A1**, which readily underwent cyclic etherification to afford **D**-type 3-alkynyl-2,5-dihydrofurnans due to the presence of the nucleophilic OH group (Figure 1b).[6] We report herein the observation of the Rh-catalyzed highly regio- and stereo-selective aerobic cross-coupling of terminal alkynes with 2,3-allenols affording the stereo-defined 1,3-enynals or enyl ketones with an exclusive *E* selectivity (Figure 1c), which are a type of common unit in natural or bioactive products (Figure 1d).[7]

METHODS

We chose the challenging terminal alkyne but-3-yn-1-ol **1a** with 1-(*p*-tolyl)-2,3-allenol **2a** to optimize the reaction conditions. After tedious screening of the reaction parameters, it was observed that in the presence of 5 mol % $[Cp^*RhCl_2]_2$ and 2.5 equiv of $Cu(OAc)_2{\cdot}H_2O$ in acetonitrile and methanol at 50 °C, the reaction did afford an enyne product, which is

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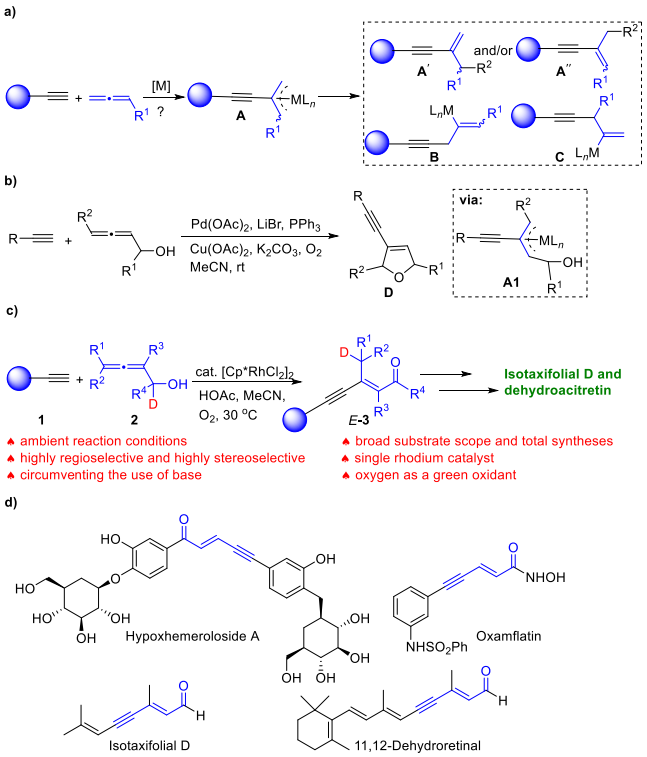


Figure 1. Selected examples of naturally occurring and bioactive enynes and terminal alkyne-based approaches to enynes. (a) Possible reactions of terminal alkynes with allenes: the selectivity issues. (b) Known cyclization of terminal alkynes with 2,3-allenols. (c) Rh(III)-catalyzed linear cross-coupling of terminal alkynes with allenols for enynyl aldehyde or ketone syntheses. (d) Natural products and bioactive molecules bearing 1,3-enynyl carbonyl units.

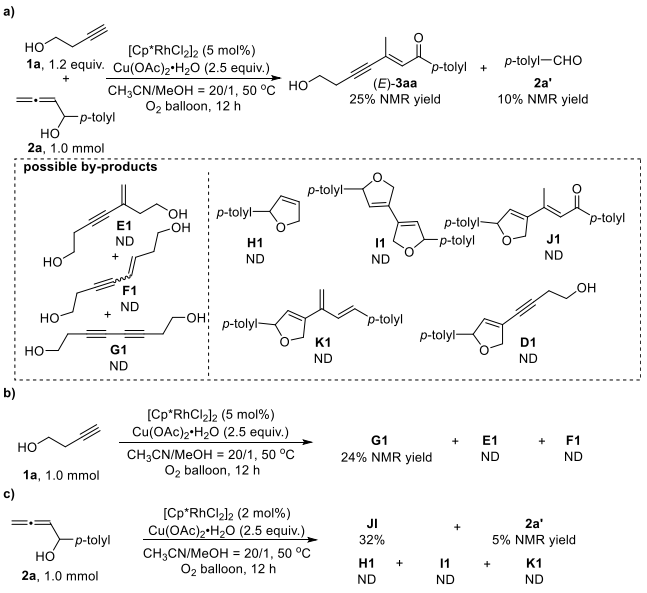


Figure 2. Rh-catalyzed reactions of terminal alkyne **1a** with 2,3-allenol **2a**, terminal alkyne **1a**, or 2,3-allenol **2a**. (a) Rh(III)-catalyzed cross-coupling of terminal alkyne **1a** with allenol **2a**. (b) Rh(III)-catalyzed reaction of terminal alkyne **1a**. (c) Rh(III)-catalyzed reaction of allenol **2a**.

different from either **A′** or **A″**, as shown in Figure 1a. It has been characterized as 3-alkynyl-(2*E*)-enone (*E*)-**3aa**. The reaction afforded (*E*)-**3aa** in 25% nuclear magnetic resonance (NMR) yield with a 10% NMR yield of allenol decomposition product of *p*-methylbenzaldehyde **2a′**. It is interesting to observe that the bimolecular regioisomeric enyne products **E1** and **F1**,[8] Glaser coupling product conjugated diyne **G1**,[9] cycloisomerization product **H1**,[10] bi-2,5-dihydrofuran product **I1**,[11] 2,5-dihydrofuryl enone product **J1**,[12] 4-(1,3-dienyl)-2,5-dihydrofuran product **K1**,[13] and 3-alkynyl-2,5-dihydrofuran product **D1**[6] were all not detected (Figure 2a). However, control experiments showed that the Glaser coupling product **G1** was formed in 24% NMR yield, and the bimolecular regioisomeric enyne products **E1** and **F1** were not formed when but-3-yn-1-ol **1a** was treated alone under the same conditions (Figure 2b); when 1-(*p*-tolyl)-2,3-allenol **2a** alone was stirred under the same conditions, 32% of 2,5-dihydrofuryl enone **J1** was detected with 5% NMR yield of *p*-methylbenzaldehyde **2a′**;[12] the formation of cycloisomerization product **H1**, bi-2,5-dihydro furan product **I1**, and 4-(1,3-dienyl)-2,5-dihydrofuran product **K1** were not observed (Figure 2c). These two sets of results indicated the perfect match for the reactivities of the terminal alkyne and 2,3-allenol.

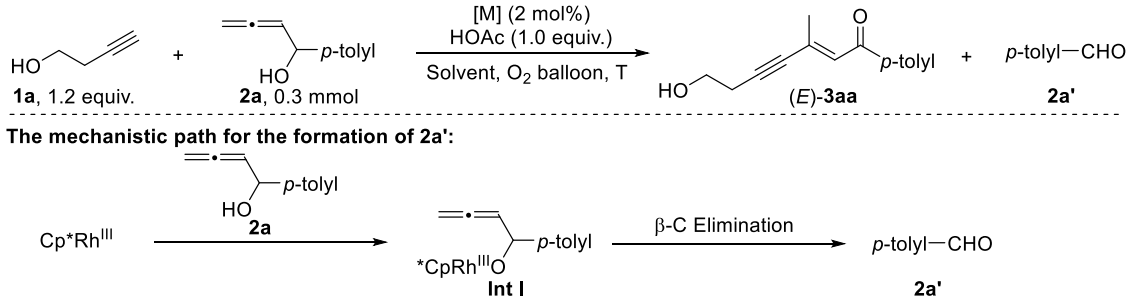
Further optimization led to the observation that the same reaction may be conducted in the absence of $Cu(OAc)_2{\cdot}H_2O$ in MeCN to afford 3-alkynyl-(2*E*)-enone (*E*)-**3aa** in 34% NMR yield stereoselectively together with 2% NMR yield of *p*-methylbenzaldehyde **2a′** with just 2 mol % $[Cp^*RhCl_2]_2$ (Table 1, entry 1). The ligand exchange of Cp^*Rh^{III} with 2,3-allenol **2a** would generate intermediate **Int I**, which would undergo subsequent *β*-carbon elimination to afford tolylbenzaldehyde **2a′**. Interestingly, the reaction in the presence of HOAc (1 equiv) improved the yield of (*E*)-**3aa** to 41% NMR yield (Table 1, entry 2). The yield of (*E*)-**3aa** dropped with 2 equiv of HOAc (Table 1, entry 3). The reaction at the ambient temperature of 30 °C afforded (*E*)-**3aa** in 51% NMR yield with 9% recovery of **2a** (Table 1, compare entry 5 with entries 2 and 4). When $Pd(OAc)_2$, $Ni(OAc)_2{\cdot}4H_2O$, or $[Cp^*IrCl_2]_2$ was applied as the catalyst, the reaction did not work (Table 1, entries 6∼8). Running the reaction at a concentration of 0.25 M for 16 h further improved the yield of (*E*)-**3aa** to 66% with a complete conversion (Table 1, compare entry 9 with entries 5 and 10). Subsequent solvent screening (Table 1, entries 11∼17) indicated that acetonitrile is still the best. The yield of (*E*)-**3aa** dropped in the presence of 10 mol % of silver salts (AgOAc, $AgNO_3$, or Ag_2CO_3) or 10 mol % of $Cu(OAc)_2{\cdot}H_2O$ (Table 1, compare entry 9 with entries 18∼21). Thus, the reaction conditions of entry 9 in Table 1 have been defined as the standard conditions for further scope study, as shown in Figure 3. The reaction enjoys single rhodium catalysis and the use of a base for the terminal alkynylmetalation is not necessary.

RESULTS AND DISCUSSION

As shown in Figure 3, terminal alkynols with different lengths of carbon chain between the C−C triple bond and the hydroxy group afforded the corresponding products (*E*)-**3aa**∼(*E*)-**3ca** in decent yields, and the reaction is robust since no protection is required for the free OH group. Different alkyl-substituted terminal alkynes without the hydroxy group also worked to afford (*E*)-**3da** and (*E*)-**3ea**. Even the highly strained cyclopropyl group survived, affording (*E*)-**3fa**. Terminal alkynes bearing various highly synthetically versatile functional groups, such as chloro (**1g**), cyano (**1h**), and ester (**1i**), may also be applied, affording (*E*)-**3ga**∼(*E*)-**3ia**. Even the reaction

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Table 1. Optimization of Reaction Conditions[a]



entry	[M]	solvent	T (°C)	time (h)	conc. (M)	NMR yield/recovery (%)[b] (*E*)-**3aa**	**2a′**	**2a**
1	$[Cp^*RhCl_2]_2$[c]	MeCN	50	15	0.5	34	2	0
2	$[Cp^*RhCl_2]_2$	MeCN	50	12	0.5	41	2	0
3	$[Cp^*RhCl_2]_2$[d]	MeCN	50	12	0.5	31	2	0
4	$[Cp^*RhCl_2]_2$	MeCN	40	11	0.5	38	1	8
5	$[Cp^*RhCl_2]_2$	MeCN	30	11	0.5	51	1	9
6	$Pd(OAc)_2$	MeCN	30	12	0.5	0	0	58
7	$Ni(OAc)_2{\cdot}4H_2O$	MeCN	30	12	0.5	0	0	100
8	$[Cp^*IrCl_2]_2$	MeCN	30	10	0.5	0	2	20
9	**$[Cp^*RhCl_2]_2$**	**MeCN**	**30**	**16**	**0.25**	**66**	**2**	**0**
10	$[Cp^*RhCl_2]_2$	MeCN	30	24	0.2	56	1	14
11	$[Cp^*RhCl_2]_2$	dioxane	30	24	0.25	5	0	44
12	$[Cp^*RhCl_2]_2$	THF	30	24	0.25	13	0	19
13	$[Cp^*RhCl_2]_2$	toluene	30	18	0.25	16	2	0
14	$[Cp^*RhCl_2]_2$	MeOH	30	13	0.25	4	0	0
15	$[Cp^*RhCl_2]_2$	DCM	30	13	0.25	37	2	0
16	$[Cp^*RhCl_2]_2$	MeCN/DCM[e]	30	29	0.25	38	1	0
17	$[Cp^*RhCl_2]_2$	MeCN/toluene[f]	30	29	0.25	58	1	1
18	$[Cp^*RhCl_2]_2$[g]	MeCN	30	16	0.25	34	0	27
19	$[Cp^*RhCl_2]_2$[h]	MeCN	30	16	0.25	25	1	33
20	$[Cp^*RhCl_2]_2$[i]	MeCN	30	16	0.25	<5	0	82
21	$[Cp^*RhCl_2]_2$[j]	MeCN	30	16	0.25	49	10	0

[a]Reaction conditions: terminal alkyne **1a** (0.36 mmol), allenol **2a** (0.3 mmol), $[Cp^*RhCl_2]_2$ (0.006 mmol), and HOAc (0.3 mmol) in CH_3CN (0.6 mL) under an O_2 atmosphere (balloon) unless otherwise noted. [b]Determined by NMR. [c]The reaction was conducted in the absence of HOAc. [d]2.0 equiv of HOAc was used. [e]The solvent of MeCN/DCM = 3/1 was used. [f]The solvent of MeCN/toluene = 3/1 was used. [g]10 mol % of AgOAc was added. [h]10 mol % of $AgNO_3$ was applied. [i]The reaction was conducted with 10 mol % of Ag_2CO_3. [j]10 mol % of $Cu(OAc)_2{\cdot}H_2O$ was added.

of *p*-formyl phenylacetylene furnished the corresponding product (*E*)-**3la** with the formyl group untouched, whose structure was further confirmed by X-ray crystallographic study.[14] Different aryl- and heteroaryl-substituted terminal alkynes **1m**∼**1o** also worked to afford the corresponding 3-alkyn-2(*E*)-enones (*E*)-**3ma**∼(*E*)-**3oa**. However, heteroaryl-substituted terminal alkyne 2-pyridinylacetylene **2p** afforded trace (*E*)-**3pa**, with 88% recovery of **2a**. When hepta-1,6-diyne was applied, the selectivity is remarkable: the mono-cross-coupling product (*E*)-**3qa** was obtained in 41% yield, together with the double-cross-coupling product (*E*)-**3qaa** in only 5% yield. With regard to 2,3-allenols, R^5 may be alkyl groups ((*E*)-**3db**, (*E*)-**3rb**, (*E*)-**3be**) or electron-rich and electron-deficient aryl groups ((*E*)-**3aa**∼(*E*)-**3qa**, (*E*)-**3dc**, (*E*)-**3sd**, (*E*)-**3bf**, and (*E*)-**3bg**). Heteroaryl-substituted 2,3-allenol **2h** also delivered the product (*E*)-**3bh**. It is worthy to mention that even the aldehydes (*E*)-**3bi** and (*E*)-**3si**∼(*E*)-**3ui** with a very high *E*/*Z* selectivity were available from the reaction of terminal alkynes **1b** and **1s**∼**1u** with 2,3-butadienol **2i**.[15] To our delight, the reaction of even *di*-substituted 2,3-allenols **2j** and **2k** with terminal alkyne **1a** also afforded the desired products (*E*)-**3aj** and (*E*)-**3ak** in 39% and 34% yields, respectively. However, when the 4-position-substituted 2,3-allenols **2l** and **2m** were treated under the standard conditions with terminal alkyne **1a**, the target products were afforded in low yields. Unfortunately, when *tri*- and *tetra*-substituted 2,3-allenols **2n**∼**2p** were treated with **1a** under the standard conditions, (*E*)-**3an**∼(*E*)-**3ap** were not formed; the reaction of *tri*-substituted 2,3-allenol **2n** with **1a** afforded 12% of the cycloisomerization product **2n″** with 81% recovery of **2n**. To demonstrate the synthetic potential of the protocol, we applied the strategy to derivatize several bioactive molecules, as shown in Figure 3; (*S*)-naproxen, a nonselective cyclooxygenase inhibitor, was transformed to the corresponding allenol (2*S*)-**2q** via reduction, aerobic oxidation, 1,2-addition, and ATA reaction,[2d] which reacted with terminal alkyne, hept-1-yne **1d**, to afford (*S*,*E*)-**3dq** in 47% yield. The esterification of pent-4-ynoic acid with L-menthol afforded terminal alkynoate (1*R*,2*S*,5*R*)-**1v**, which was treated with allenol **2a** to afford enynone (1*R*,2*S*,5*R*,*E*)-**3va** in 52% yield. A similar operation was also conducted with (*S*)-ibuprofen, a nonsteroidal anti-inflammatory drug, affording the product (*S*,*E*)-**3wa** in a 47% yield. Terminal propargylic alcohol (5*S*)-**1x** was synthesized from (−)-citronellal. Subsequent ATA reaction[2d] afforded allenol (6*S*)-**2r**, which was easily transformed to (*S*,*E*)-**3dr** under the standard conditions by its treatment with terminal alkyne of hept-1-yne **1d**. The reaction

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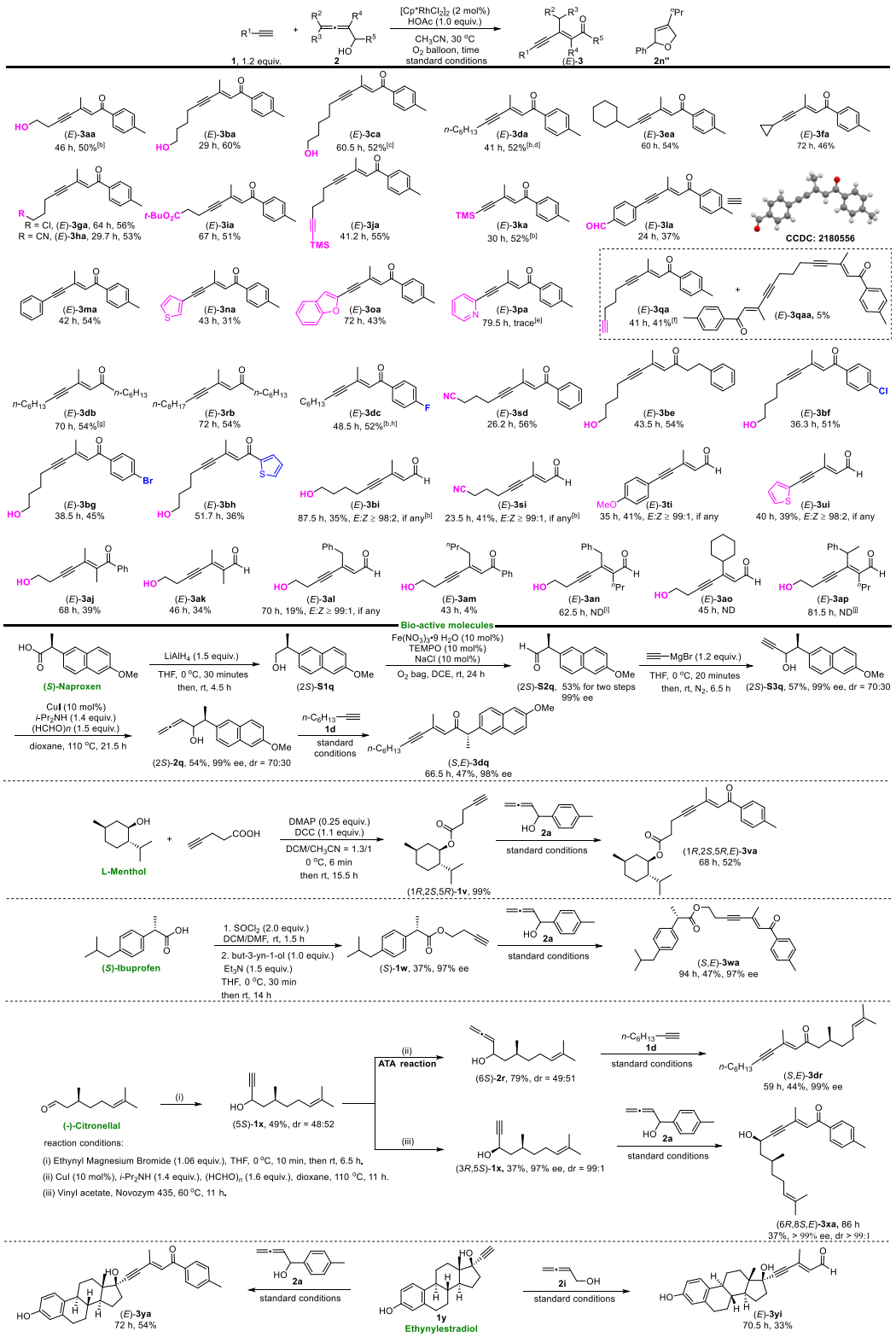


Figure 3. Substrate scope. (a) Reaction conditions: terminal alkyne **1** (0.6 mmol), allenol **2** (0.5 mmol), $[Cp^*RhCl_2]_2$ (0.01 mmol), and HOAc (0.5 mmol) in CH_3CN (2 mL) at 30 °C under an O_2 atmosphere (balloon) unless otherwise noted. (b) 1.0 mmol scale. (c) 4% recovery of **2a** was detected by NMR. (d) 5% recovery of **2a** was detected by NMR. (e) 88% recovery of **2a** was detected by NMR. (f) 2.8 equiv of **1q** was used. (g) 6% recovery of **2b** was detected by NMR. (h) 9% recovery of **2c** was detected by NMR. (i) 12% of **2n″** was obtained by isolation; 81% recovery of **2n** was detected by NMR. (j) 83% recovery of **2p** was detected by NMR.

of optically active propargylic alcohol (3*R*,5*S*)-**1x**, which was readily available from enzymatic separation catalyzed by Novozym 435,[16] with allenol **2a** afforded (6*R*,8*S*,*E*)-**3xa** in 37% yield. Commercially available ethinylestradiol **1y** may be converted to product (*E*)-**3ya** in 54% yield by its reaction with secondary 2,3-allenol **2a** under the optimized reaction

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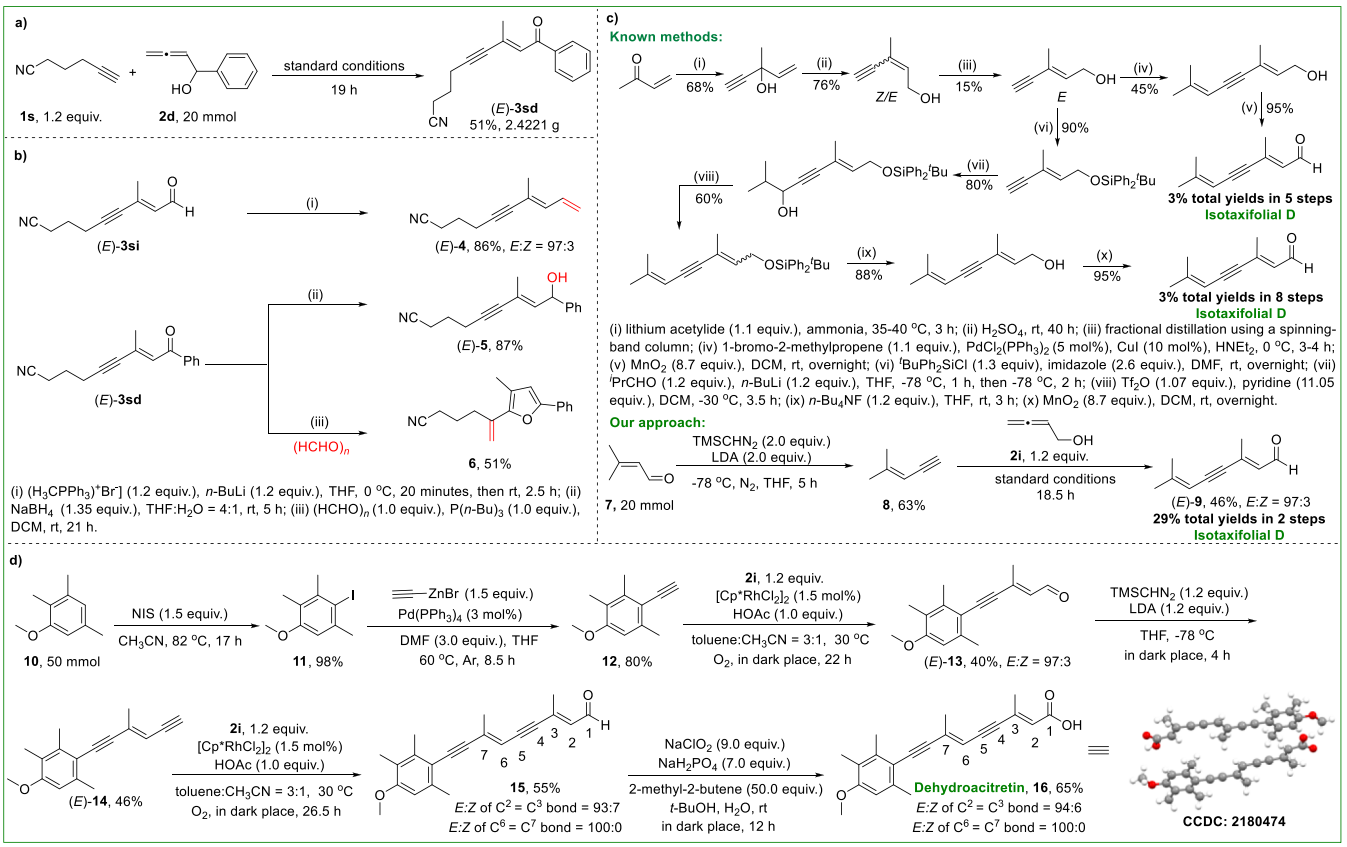


Figure 4. Gram-scale reaction and synthetic applications. (a) Gram-scale reaction. (b) Synthetic applications. (c) Total synthesis of isotaxifolial D. (d) Synthesis of dehydroacitretin.

conditions. Even the reaction of 2,3-butadienol **2i** with ethinylestradiol **1y** afforded the aldehyde product (*E*)-**3yi** in 33% yield.

To further explore the synthetic potentials of this transformation, a gram-scale synthesis of (*E*)-**3sd** was executed under the standard reaction conditions (Figure 4a); treatment of alkynenal (*E*)-**3si** with a Wittig reagent afforded conjugated alka-1,3-dien-5-yne (*E*)-**4** in 86% yield;[17] controlled reduction of the ketone functionality without touching the cyano functionality in (*E*)-**3sd** with $NaBH_4$ resulted in secondary alcohol (*E*)-**5** in 87% yield; phosphine initiated cyclization of (*E*)-**3sd** to furnish α-vinylfuran bearing a cyano group **6** in the presence of paraformaldehyde (Figure 4b).[18] This methodology has been applied to deliver a highly efficient stereoselective entry to the natural product isotaxifolial D (Figure 4c): commercially available 3-methylbut-2-enal **7** reacted with $TMSCHN_2$ to afford 4-methylbut-3-en-1-yne **8** in 63% yield,[19] which underwent this rhodium-catalyzed coupling reaction with 2,3-butadienol **2i** to afford isotaxifolial D (*E*)-**9** efficiently in 46% yield and an *E*/*Z* selectivity of 97:3. We obtained isotaxifolial D with a total yield of 29% in 2 steps. As a comparison, the known methods required 5 or 8 steps to obtain isotaxifolial D in 3% total yields from commercially available methyl vinyl ketone via the problematic separation of the *E* and *Z* isomers of 3-methylpent-4-yn-2-enol as the third step.[7b,20,21] Commercially available 2,3,5-trimethylanisole **10** was treated with NIS to afford aryl iodide **11** in 98% yield,[22] which underwent the Negishi coupling to yield terminal aryl acetylene **12**.[23] Aldehyde (*E*)-**13** was formed in 40% yield with 97:3 *E*/*Z* selectivity by this rhodium-catalyzed coupling reaction with 2,3-butadienol **2i**. (*E*)-**13** reacted with $TMSCHN_2$ in the presence of LDA to afford terminal alkyne (*E*)-**14** in 46% yield.[19] This Rh(III)-catalyzed coupling reaction of (*E*)-**14** with 2,3-butadienol **2i** was applied for the second time to afford aldehyde **15** in 55% yield with a decent stereoselectivity of 93:7 for the C3−C2 double bond. By following Lindgren−Kraus−Pinick oxidation,[24] the stereoselective synthesis of dehydroacitretin **16**[25] has been realized in 65% yield with an *E*/*Z* ratio of 94:6. The structure of product **16**[26] was confirmed by single-crystal X-ray diffraction analysis.

To gain insight into the nature of the cross-coupling reaction, some control experiments were conducted (Figure 5a,b): firstly, the same reaction was conducted under nitrogen, pure oxygen, and air atmosphere, and the results indicated that oxygen was critical for this transformation; when deuterated terminal alkyne [D]-**1r** was applied for its reaction with allenol **2b** under the standard conditions, no deuterium incorporation in the product (*E*)-**3rb** was observed; interestingly, the reaction of 1-decyne **1r** with deuterated allenol [D]-**2b** afforded [D]-(*E*)-**3rb** with the deuterium incorporation of 99% in the methyl group connected to the C−C bond. Based on these experimental data, a possible mechanism has been proposed, as shown below using **1r** and [D]-**2b** as the model substrates (Figure 5c): firstly, alkynyl rhodium intermediate **IN-1** was readily generated from the deprotometalation of the terminal alkyne **1r**, which underwent the regio- and hydroxy group-dictated stereo-selective insertion with 2,3-allenol efficiently due to the high reactivity of allene to generate 2-alkynyl π-allylic rhodium intermediate **IN-2**.[27] It would undergo C−C bond rotation, intramolecular ligand exchange, and subsequent β−D elimination forming the C−O bond

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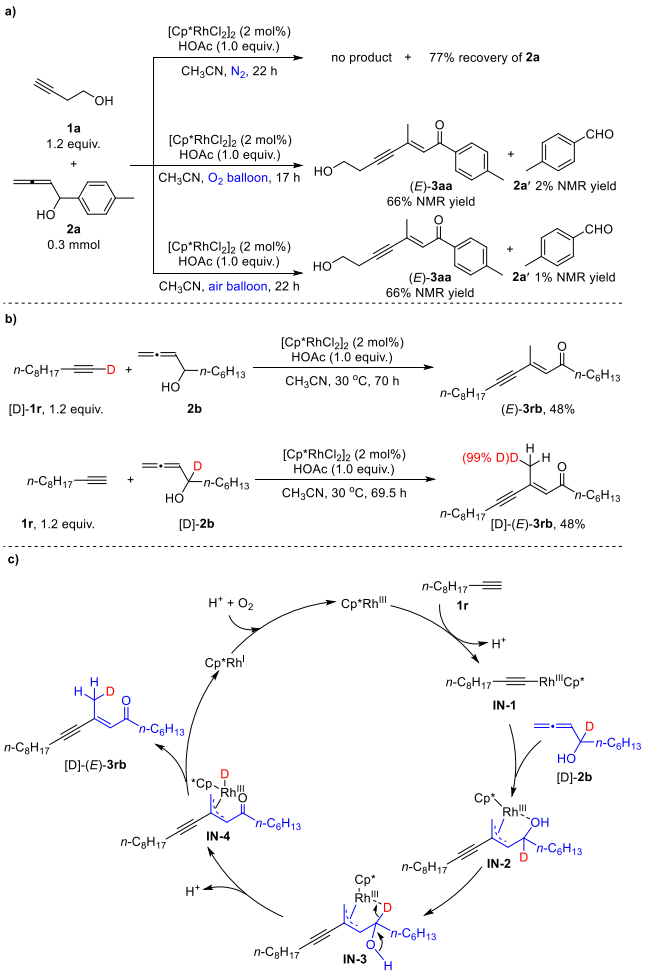


Figure 5. Mechanistic exploration and the proposed mechanism. (a) Control experiments. (b) Deuterium labeling experiments. (c) Proposed mechanism.

resulting in the stereodefined π-allylic rhodium hydride **IN-4** via the intermediacy of **IN-3**. This formal 1,4-D delivery was achieved via subsequent reductive elimination to yield the final product [D]-(*E*)-**3rb** and Rh(I) species. Under the oxygen atmosphere and proton, Rh(III) was regenerated to finish the catalytic circle.

CONCLUSIONS

In summary, we have developed a conceptually new approach for the enyne synthesis via a single rhodium-catalyzed cross-coupling reaction of terminal alkynes with 2,3-allenols. This protocol enables an efficient and highly selective preparation of (*E*)-3-alkynyl enones/enals under a set of ambient conditions, which could serve as platform molecules for the synthesis of many other useful molecules due to the accommodation of a wide array of highly reactive functional groups. The synthetic potential of this methodology has been demonstrated via derivatization of bio-active molecules, the efficient total syntheses of isotaxifolial D, and the synthesis of dehydroacitretin. A mechanism has been proposed for the observed highly selective formation of (*E*)-3-alkynyl enones/enals. The use of a base is not required since the terminal alkynylmetalation proceeds readily under base-free conditions, and the catalytically active rhodium catalyst has been regenerated via aerobic oxidation with the help of HOAc. The development of many other types of such reactions with both types of readily available terminal alkynes and allenes as the reactants for enyne syntheses is highly expected and is being actively pursued in our laboratory.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c01327.

General information; synthesis of starting materials; Rh-catalyzed reactions of terminal alkyne **1a** with 2,3-allenol **2a**, terminal alkyne **1a**, or 2,3-allenol **2a**; Rh-catalyzed aerobic E-selective linear coupling of terminal alkynes with 2,3-allenols; gram-scale reaction and synthetic applications; mechanistic studies; and NMR and HPLC spectra of the compounds prepared (PDF)

Crystallographic data for product (*E*)-**3la** (CIF)

Crystallographic data for product **16** (CIF)

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Author Contributions

§Y.S. and X.W. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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